

Mathematics Detailed Practice Workbook

Concepts • Problems • Mini Mock

Part 1 — Foundation

- Find the 20th term of an AP with $a=5$ and $d=3$.
- Find the number of ways to choose 3 objects from 8 distinct objects.
- Expand the first few terms of $(x+2)^4$.
- Find the determinant of a 2×2 matrix with entries a, b, c, d .
- Find the distance between two given coordinate points.

Part 2 — Calculus

- Evaluate a standard trigonometric limit after suitable simplification.
- Differentiate $y=(x^2+1)^5$.
- Find the stationary points of a simple polynomial function.
- Evaluate an indefinite integral using substitution.
- Use a definite-integral property to reduce the amount of calculation.

Part 3 — Vectors & 3D

- Find the magnitude of a vector with given components.
- Check whether two vectors are perpendicular.
- Find the area of a triangle using a vector cross product.
- Write direction ratios for a line through two 3D points.
- Explain the geometric meaning of a dot product.

Part 4 — Probability

- A fair die is thrown twice. Define events and find the probability of a stated condition.
- Two events have given probabilities and intersection. Find their union probability.
- Use conditional probability in a simple selection problem.
- Determine whether two events are independent from their probabilities.
- Find expected value for a small discrete random variable.

Mini Mock

1. The discriminant of a quadratic with two equal roots is: A) positive B) zero C) negative D) one.
2. nCr represents: A) arrangement B) selection C) derivative D) determinant.
3. $d(x^2)/dx$ equals: A) x B) $2x$ C) x^2 D) 2.

- 4. If $a \cdot b = 0$, the non-zero vectors are: A) parallel B) perpendicular C) equal D) opposite only.
- 5. $P(A|B)$ equals: A) $P(A)/P(B)$ B) $P(A \cap B)/P(B)$ C) $P(A \cup B)$ D) $P(B)/P(A)$.
- Answers: 1-B, 2-B, 3-B, 4-B, 5-B.